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Mapping the Structure and Thematic Landscape of Latin American Engineering Journals Indexed in SciELO and Scopus

Mapeo de la estructura y el panorama temático de las revistas latinoamericanas de ingeniería indexadas en SciELO y Scopus

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Abstract
This study presents a bibliometric and network-based analysis of engineering journals published in Latin America and indexed in both SciELO and Scopus simultaneously. It aims to characterize their disciplinary scope, thematic specialization, and scientific collaboration. Using a correlation map of Scopus subject areas and performance indicators, the study identifies key contributors, highlighting the roles of leading institutions and the global visibility of top journals. Author keywords reveal thematic trends, terminological consistency, and a shift toward high-impact areas, including digital technologies, artificial intelligence, and sustainability. Inter-journal relationships are explored through keyword correlation, while citation, co-citation, and bibliographic coupling networks highlight intellectual linkages. A co-occurrence analysis identifies dominant and emerging research themes. These include sustainability, technological innovation, public health, education, and core fields like materials, optimization, and risk analysis. The findings offer a multidimensional perspective on Latin American engineering journals and their integration into the global scholarly landscape, and provide practical insights for authors, editors, and policymakers seeking to strengthen regional engineering research.

Keywords: Engineering Journals; Bibliometric Analysis; Citation Networks; Thematic Mapping; Scientific Collaboration; Research Performance.

Resumen
Este estudio presenta un análisis bibliométrico y basado en redes de las revistas de ingeniería publicadas en América Latina e indexadas simultáneamente en SciELO y Scopus. El objetivo es caracterizar su alcance disciplinar, su especialización temática y sus patrones de colaboración científica. A través de un mapa de correlación entre áreas temáticas de Scopus y indicadores de desempeño, se identifican los principales actores, destacando el papel de instituciones líderes y la visibilidad internacional de las revistas más reconocidas. El análisis de palabras clave del autor permite identificar tendencias temáticas, consistencia terminológica y una transición hacia áreas de alto impacto, como las tecnologías digitales, la inteligencia artificial y la sostenibilidad. Las relaciones entre revistas se exploran mediante la correlación de palabras clave, mientras que las redes de citación, co-citación y acoplamiento bibliográfico evidencian vínculos intelectuales. Un análisis de coocurrencia permite identificar temas dominantes y clústeres emergentes en sostenibilidad, innovación tecnológica, salud pública, educación y áreas fundamentales como materiales, optimización y análisis de riesgos. Los hallazgos ofrecen una perspectiva multidimensional sobre las revistas latinoamericanas de ingeniería y su integración en el panorama académico global, además de brindar orientaciones prácticas para autores, editores y responsables de políticas comprometidos con el fortalecimiento de la investigación en ingeniería en la región.

Palabras clave: Revistas de Ingeniería; Análisis Bibliométrico; Redes de Citaciones; Mapeo Temático; Colaboración Científica; Desempeño Investigativo.

1 Introduction

The Scientific Electronic Library Online (SciELO) is a cooperative, open-access publishing platform launched in 1998 to enhance the visibility, accessibility, and quality of scientific journals from Latin America and other regions in the Global South [1], [2], [3], [4]. Developed by BIREME/PAHO and FAPESP, SciELO provides a robust infrastructure for indexing, publishing, and disseminating peer-reviewed scholarly content, with a strong emphasis on regional integration and multilingual dissemination [5]. It is vital to democratize access to scientific knowledge, particularly for researchers, institutions, and countries that are often underrepresented in mainstream citation databases, such as Web of Science and Scopus.

Today, SciELO includes over a thousand journals across disciplines and has become a strategic platform for fostering scientific communication in Latin America. Its alignment with global indexing standards has increased international visibility, with many SciELO journals now also indexed in Scopus [6]. Understanding the interrelationships between these journals offers valuable insights into regional collaboration, thematic specialization, and knowledge flows within and beyond Latin America.

Despite SciELO's recognized importance in promoting regional scientific visibility, there is limited comprehensive research on the structural and thematic interrelationships among Latin American journals indexed in both SciELO and Scopus, particularly in engineering.

This study aims to fill that gap by providing an in-depth bibliometric and network-based analysis of engineering journals published in Latin America and indexed in both databases. The analysis begins by examining the disciplinary breadth of these journals through a cross-correlation map of Scopus subject areas assigned at the journal level, highlighting the distribution and overlap of thematic fields. Fundamental performance indicators—such as publication volume and citation impact—are used to assess authors, institutions, countries, and journals and identify leading contributors. Furthermore, author keywords are analyzed to explore thematic patterns and terminological consistency. A correlation map based on shared author keywords analyzes the interrelationships among journals, where stronger connections imply greater thematic similarity. In addition, a citation network reveals the direct referencing structure among journals, while co-citation and bibliographic coupling analyses uncover indirect intellectual linkages. Finally, a co-occurrence network of author keywords is used to identify dominant research themes and conceptual clusters within the engineering literature. Together, these analyses provide a multidimensional view of the structure, dynamics, and thematic orientation of engineering research in Latin America, offering valuable insights into regional integration, collaboration, and visibility in the global scientific landscape.

This study focuses exclusively on engineering journals published in Latin America and indexed in both SciELO and Scopus. By limiting the analysis to journals present in both databases, the study ensures international visibility and metadata consistency, facilitating bibliometric and network-based comparisons. This scope excludes engineering journals that are indexed in only one of the two databases. The analysis is conducted at the journal level, rather than at the level of individual authorship networks, which may limit the granularity of insights into micro-level collaboration patterns. Ultimately, the study offers a snapshot of the structural and thematic dynamics based on the available metadata at the time of analysis. It does not account for recent changes or future trends in journal indexing policies or editorial practices.

The rest of this paper is organized as follows: Section 2 discusses the methodology used. Section 3 presents the results. Section 4 discusses the findings. Finally, Section 5 presents the conclusions.

2 Materials and Methods

This section outlines and examines the standard workflow commonly employed in literature analysis and tech-mining studies, which also serves as the methodological foundation of this paper. This process has been widely discussed in the research literature, including by Aria and Cuccurullo [7], Donthu et al [8], Page et al [9], and Zupic and Carter [10]. The adopted methodology comprises four key stages:

1. Design of the study.
2. Collection and preparation of data.
3. Analysis of the data.
4. Interpretation of the findings.

2.1 Design of the Study

The methodological parameters of this study are presented below, following the literature review criteria established by [11], [12]. The search strategy consisted of identifying the ISSNs of relevant journals in SciELO (and indexed in Scopus) and using them to conduct targeted searches in Scopus. The study parameters are:

- Database: Scopus.
- Years of analysis: From Jan 2015 to Dec 2024.
- Data retrieval: Feb 3, 2025.
- Search string: The ISSN search operator was used to retrieve information on each engineering journal indexed in SciELO from Scopus.
- Inclusion criteria: None.
- Exclusion criteria: Documents whose abstracts are not written in English.

The exclusion criterion was based on two primary considerations: the predominance of English as the language of abstracts in the selected journals and the use of text-processing tools optimized for English-language content. Consequently, documents with abstracts in languages other than English were excluded to ensure consistency and analytical accuracy.

Fig. 1 presents the PRISMA flow chart, a standardized framework for reporting the identification, screening, eligibility, and inclusion of documents in systematic reviews and meta-analyses. During the identification phase, the search string returned 29,499 papers. In the screening phase, a time filter was applied, excluding 12,849 papers published before 2015 or in 2025. The eligibility phase involved reviewing the titles and abstracts of the remaining 16,603 documents published between 2015 and 2024, excluding 1,307 papers. As a result, the final dataset consists of 15,286 documents.

2.2 Collection and preparation of data

Following widely accepted practices in the literature, all bibliographic data were downloaded from Scopus in CSV format for analysis [8], [9]. The downloaded dataset includes document titles, abstracts, author keywords, index keywords, authors, affiliations, source titles, and bibliographic references.

Data processing combined automated procedures with manual refinements to ensure consistency and quality. The cleaning process applied to the raw author keywords included:

- Converting all text to uppercase.
- Translating British spelling to American variants.
- Removing multiple consecutive spaces within strings.
- Standardizing the formatting of hyphenated words.

To unify text strings referring to the same concept, we applied rule-based normalization that grouped equivalent terms via exact-match merging, applied stemming patterns, and manually inspected ambiguous cases to ensure conceptual consistency across author keywords. This preprocessing step was essential to ensure accurate keyword grouping and analysis in the subsequent stages. All preprocessing, keyword cleaning, and network analyses were performed entirely in Python using the TechMiner package, with no external software required for visualization or data processing. Author keywords were used directly for thematic exploration, reflecting the terminology and concepts the authors prioritized. Identifying dominant themes based on cleaned author keywords supports a multifaceted analysis of the engineering papers published in SciELO, as detailed in the following section.

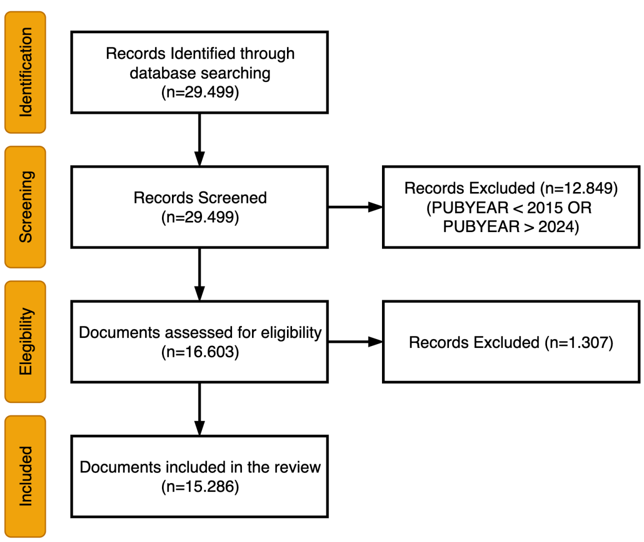

Figure 1. The PRISMA flow chart
Source: The authors.

2.3 *Analysis of the data*

Various analyses were conducted to understand better the papers published in engineering journals indexed in SciELO. The following sections detail the specific procedures and findings of each study.

2.4 *Interpretation of the findings*

The interpretation phase focused on reviewing and synthesizing results from the various analyses to draw conclusions aligned with this study's objectives. Patterns, trends, and thematic clusters identified during the data analysis phase were examined in depth and contextualized within the broader landscape of engineering research in Latin America. This step ensured that the insights were coherent and relevant to the study's overall analytical framework.

3 Results and Discussion

This section presents the fundamental bibliometric indicators of the analyzed dataset.

3.1 *General Metrics*

The dataset includes scientific publications from January 2015 to December 2024, totaling 15,286 documents and reflecting an annual growth rate of 28.05%. The average document age is 5.77 years, with each work receiving approximately 4.11 citations overall, or 0.41 citations per year. Publications originate from 37 different sources, averaging 413.14 documents per source. The dataset comprises 14,822 journal articles, 35 conference papers, 21 editorials, two errata, one letter to the editor, and 402 review papers. A total of 40,773 authors (1,609 unique) contributed to the publications, with a strong tendency toward collaboration, averaging 3.61 authors and 3.92 co-authors per document. International collaborations account for 18.84% of the dataset. The contributing authors are affiliated with 10,687 organizations across 127 countries.

These patterns are consistent with findings from bibliometric studies in other disciplines—such as health sciences, management, and education—where thematic clustering and institutional concentration similarly influence the structure and evolution of scholarly production.

3.2 *Leading Scopus Subject Areas*

This section analyzes the subject areas assigned by Scopus, which are attributed at the journal level rather than to individual articles. These classifications help identify the disciplinary focus of the research published in SciELO-indexed engineering journals. The dataset used in this study comprises 15,286 documents distributed across 18 of Scopus's 27 subject areas. Notably, eight subject areas are associated with 1,000 or more documents. The most prominent areas are:

- Engineering: 7,257 documents (as anticipated).
- Materials Science: 2,664 documents.
- Computer Science: 1,884 papers.
- Chemical Engineering: 1,629 documents.
- General: 1,609 documents.
- Agricultural and Biological Sciences: 1,401 papers.
- Environmental Science: 1,263 documents.
- Business, Management, and Accounting: 1,017 papers.

Fig. 2 displays a cross-correlation map between Scopus subject areas and SciELO journals. This visualization provides insight into the interconnections between journals based on their associated subject areas. The numbers alongside each subject area indicate the corresponding documents and citations. Links between nodes reflect the degree of similarity, while node size is proportional to the number of documents associated with each subject area.

The correlation map reveals a strong central cluster dominated by Engineering and its closely related subfields, such as Materials Science, Computer Science, and Energy, reflecting the core disciplinary focus of the journal set. However, the map also highlights additional groups of subject areas that, while internally cohesive, exhibit weaker connections to the engineering core. For example, Physics and Astronomy and Mathematics form a distinct, well-defined cluster, whereas Medicine, Immunology and Microbiology, Biochemistry, Genetics and Molecular Biology, and Environmental Science are tightly interrelated but display limited thematic overlap with engineering. Notably, the Social Sciences appear as an isolated node, indicating minimal correlation with other disciplines in the network. These peripheral clusters underscore the multidisciplinary intersections in the dataset while reinforcing the centrality of engineering within the overall structure.

3.3 Basic Performance Metrics

3.3.1 Authors

Table 1 presents the 20 most frequent authors publishing in engineering journals indexed in SciELO between 2015 and 2024, including the number of papers, citations received within SciELO, total publications indexed in Scopus, and the ratio of SciELO to Scopus publications. This ratio provides insight into each author's publication preferences and engagement with the SciELO platform in the broader Scopus-indexed literature.

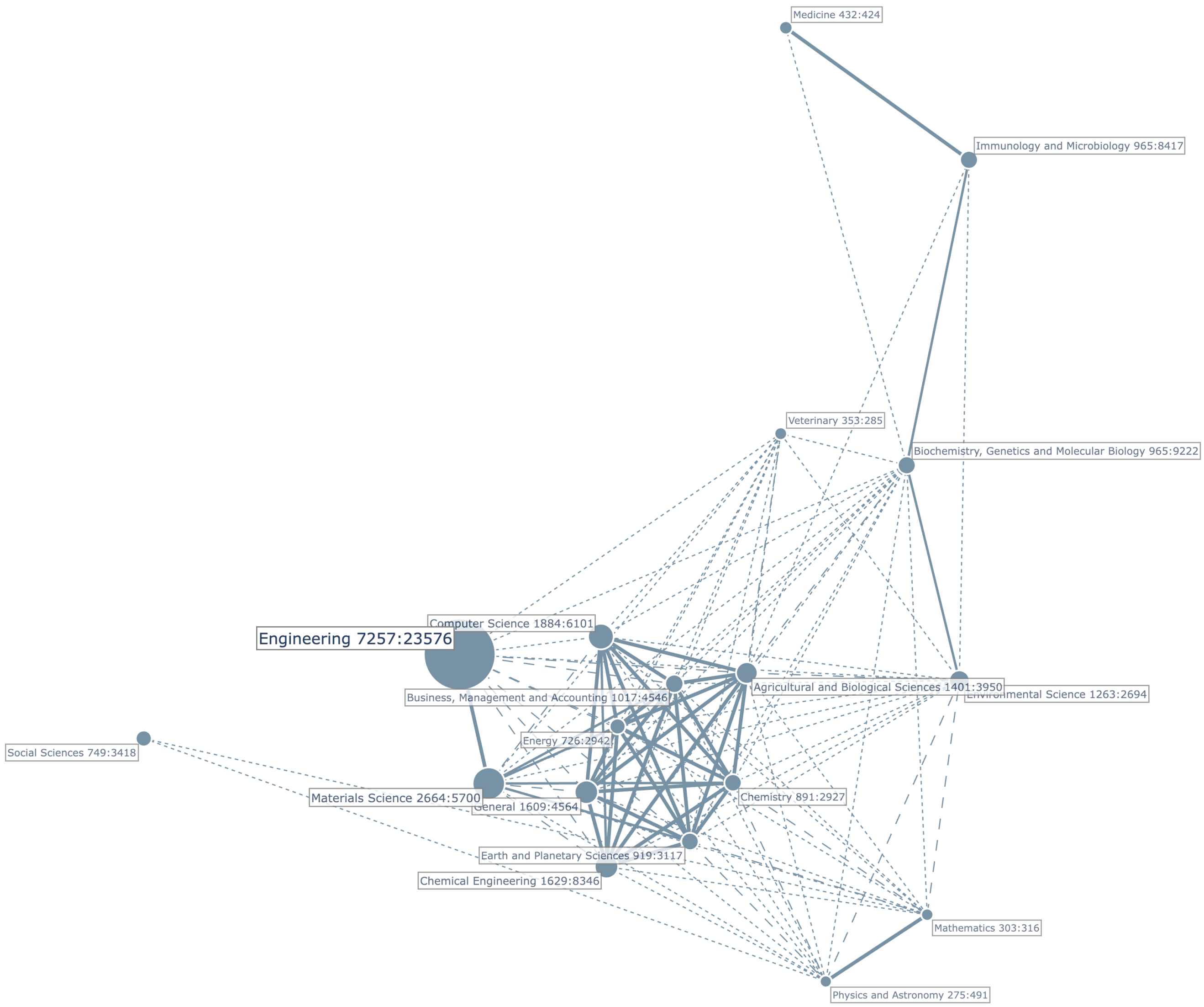

Figure 2. Correlation map of Scopus Subject Areas crossed with engineering SciELO journals.
Source: The authors.

Table 1.
Most Frequent Authors.

Author Name	Papers SciELO	Citations Scielo	Papers Scopus	Ratio
Ganga-Contreras F.A.	39	182	153	25.4 %
Pedraja-Rejas, L.M.	37	240	108	34.3 %
Campos-Aranda D.F.	37	64	43	86.0 %
Severino-González P.	36	221	61	59.0 %
Gelbukh A.	33	219	287	11.5 %
Acevedo-Correa D.	33	164	57	57.9 %
López-Lezama J.M.	30	177	131	22.9 %
Vidal-Silva C.L.	29	152	88	33.0 %
Valencia-Arias A.	28	178	237	11.8 %
Muñoz-Galeano, N.	27	148	85	31.8 %
Rodríguez-Ponce E.R.	26	185	87	29.9 %
Ortega-Toro R.	26	157	96	27.1 %
Dávila-Morán, R.C.	25	21	84	29.8 %
Preciado-Rangel, P.	24	63	118	20.3 %
Escobar D.A.	23	86	88	26.1 %
Paz-Pellat F.	23	58	44	52.3 %
Bustamante-Ubilla M.A.	22	117	42	52.4 %
Fontalvo-Herrera, T.	21	167	46	45.7 %
Castrillón O.D.	20	110	33	60.6 %
Zapata J.E.	20	109	48	41.7 %

Source: The authors.

Table 2.
Most Frequent Institutions.

Name	Papers	Global Citations	Local Citations
Univ Nac de Colombia (COL)	829	3754	28
Inst Politec Nac (MEX)	602	2342	39
Univ de Antioquia (COL)	353	1406	6
Univ Nac Autonoma de Mexico (MEX)	254	1171	13
Univ Autonoma Metropolitana (MEX)	248	910	18
Univ de Tarapaca (CHL)	237	761	4
Tecnol Nac de Mexico (MEX)	234	695	7
Univ del Valle (COL)	232	983	9
CONICET (ARG)	192	528	13
Univ Ind de Santander (COL)	179	656	4

Source: The authors.

Among the authors, Ganga-Contreras F.A. leads with 39 publications in SciELO, accounting for 25.4% of their total output in Scopus, followed closely by Pedraja-Rejas, L.M., and Campos-Aranda D.F., both with 37 papers. Notably, Campos-Aranda D.F. shows a very high SciELO/Scopus ratio of 86.0%, suggesting a strong focus on SciELO as their primary publication venue. Similarly, Severino-González P. and Acevedo-Correa D. also exhibit high ratios (59.0% and 57.9%, respectively), indicating a substantial portion of their research is disseminated through SciELO journals.

In contrast, authors such as Gelbukh, A., and Valencia-Arias, A. have much lower ratios (11.5% and 11.8%, respectively), suggesting a broader, more international publication strategy that favors Scopus journals over SciELO. Despite this, they maintain a notable presence in SciELO and receive many citations.

Authors like Bustamante-Ubilla M.A. (52.4%) and Castrillón O.D. (60.6%) also strongly engage with SciELO relative to their total output. In contrast, others, such as Preciado-Rangel, P., and Escobar, D.A., have lower ratios but remain among the most frequent contributors.

3.3.2 Organizations

Table 2 lists the ten most frequent institutions contributing to engineering journals indexed in SciELO between 2015 and 2024. The total number of publications in the dataset ranks these institutions. Two citation metrics are included: Global citations, which count the total number of citations recorded in Scopus, and Local citations, which count citations made within the analyzed database.

Of the 10,687 unique organizations represented in the dataset (4,956 of which appear as first-author affiliations), the top 10 institutions account for a substantial share of the overall scientific output. These institutions are geographically concentrated in Colombia and Mexico, with eight out of the ten based in these two countries. This concentration highlights their significant contribution to regional engineering research. Notably, institutions such as the Universidad de Tarapacá (Chile) and CONICET (Argentina) also appear among the top contributors, reflecting the broader participation of Latin American countries in SciELO's indexed engineering literature.

Leading the list is the Universidad Nacional de Colombia with 829 publications, followed by the Instituto Politécnico Nacional (Mexico) with 602 publications, and the Universidad de Antioquia (Colombia) with 353 publications. Together, these three institutions account for over 11% of all documents analyzed, highlighting their central role in the engineering research landscape in SciELO.

In terms of global impact—measured by Scopus citations—the Universidad Nacional de Colombia again stands out, with 3,754 citations —the highest among all institutions —signaling strong visibility and influence beyond the local context. The Instituto Politécnico Nacional also demonstrates a high global impact, with 2,342 citations. At the same time, institutions such as the Universidad Nacional Autónoma de México and the Universidad del Valle show solid performance, with over 900 citations each.

As expected in a dataset covering a wide range of journals and topics, local citations—used here as a proxy for intra-database scholarly exchange—are generally lower. However, the Instituto Politécnico Nacional exhibits a relatively high local citation count (39), suggesting a strong internal connection within the SciELO engineering network. Other institutions, such as the Universidad Nacional Autónoma de México and CONICET, also show moderate levels of local citation activity (13 each), indicating some degree of scholarly interaction within the indexed journals.

3.3.3 Countries

Table 3 highlights the geographic distribution of contributions to engineering publications in SciELO between 2015 and 2024, underscoring the platform's role as a key hub for regional and international scholarly communication. Overall, the Americas dominate the dataset with 11,235 documents and 40,220 global citations, followed by Asia with 3,237 papers and Europe with 1,922. Within the Americas, Latin America and the Caribbean emerge as the core of SciELO's engineering output, contributing 11,094 documents, 39,040 global citations, and 506 local citations. This regional concentration reinforces SciELO's identity as a platform rooted in Latin American scholarship while also reflecting growing global engagement, particularly from Asia and Europe.

A closer look at subregional dynamics reveals that Eastern and Southern Asia show notable levels of participation, with Southern Asia standing out for its relatively high rate of local citations, suggesting meaningful engagement with the SciELO research network. Spain, though publishing fewer papers, distinguishes itself by a substantial citation impact, indicating visibility and influence within the global scientific community.

At the national level, Colombia and Mexico lead in publication volume, with 3,524 and 3,452 papers, respectively, together accounting for nearly 46% of the total 15,286 documents in the dataset. While Colombia registers the highest number of publications, Mexico demonstrates more robust intra-database citation activity, pointing to greater local scholarly interaction. China ranks third in productivity, with 2,148 papers. Although it exhibits high global citation counts, its local engagement appears limited, possibly reflecting a more external orientation in citation behavior.

Table 3.
Most Frequent Countries.

Name	Papers	Global Citations	Local Citations
Colombia	3,524	13,447	102
Mexico	3,452	12,408	154
China	2,148	6,326	59
Chile	1,265	6,100	85
Spain	1,226	5,341	42
Brazil	951	4,351	133
Ecuador	859	1,697	8
Peru	661	1,474	25
Argentina	543	1,162	15
Cuba	415	889	8

Source: The authors.

Other Latin American countries, including Chile, Brazil, Ecuador, Peru, Argentina, and Cuba, significantly contribute to the dataset, further consolidating SciELO's regional character. Their participation highlights both the collaborative potential and diversity of engineering research across Latin America.

3.3.4 Sources

Table 4 presents the ten most frequent journals in the dataset, drawn from 37 sources, with an average of 413 documents per journal. BOLETIN TECNICO leads with 1,820 papers but shows relatively low citation counts (2,810 global and 12 local), suggesting high productivity but limited impact. In contrast, DYNA, with 1,341 papers, has a much higher global citation count (5,321), indicating greater visibility. COMPT SIST and REV MEX ING QUIMICA also show strong performance, with over 1,000 and 900 papers, respectively, and significant local citations—particularly the latter, which leads in local citations (197), suggesting strong engagement within the SciELO community.

Table 4.
Most Frequent Journals.

Name	Papers	Global Citations	Local Citations
Boletin Tecnico	1,820	2,810	12
DYNA	1,341	5,321	45
Comput. Sist.	1,037	3,310	100
Rev. Mex. Ing. Quimica	903	5,404	197
Inf. Tecnol.	882	3,698	7
Interciencia	871	2,342	8
Form.Univ.	701	4,205	7
Tecnologia Ciencias Agua	625	1,552	2
Electron. J. Biotechnol.	593	11,229	85
J App Res Tech	580	5,889	65

Source: The authors.

ELECTRON J BIOTECH stands out with the highest global citation count (11,229) despite ranking lower in document volume (593 papers), reflecting a high citation impact per paper. Similarly, J APP RES TECH shows strong citation metrics relative to its output. These results suggest that while some journals prioritize volume, others exert greater influence through their citation impact. Overall, the data reflect diverse publication strategies across journals in SciELO's engineering collection, with varying balances between productivity and scholarly impact.

3.4 Temporal trends in engineering research topics

The analyzed dataset contains 42,141 unique text strings representing the author-assigned keywords. As described in Section 3.2, these raw keywords underwent a cleaning procedure to unify different text strings that represented the same concept or idea. As a result of this process, 39,450 cleaned author keywords were obtained.

Fig. 3 presents the six most frequent author keywords per year from 2015 to 2024, based on their occurrence and citation impact, and serves as the basis for analyzing temporal keyword dominance. The early years (2015–2016) featured specialized terms such as *social networking*, *bi-level programming*, and *asphaltenes*, reflecting narrower research focuses. From 2017 onward, broader and high-impact terms such as *big data*, *data mining*, and *genetic algorithms* emerged, marking a shift toward digital technologies and computational methods. Keywords such as *simulations*, *artificial neural networks*, and *optimization* have gained sustained relevance and high global citations. Education-related terms, such as *teaching*, *learning*, and *higher education*, also gained visibility, peaking around 2018-2020. More recently, terms like *machine learning*, *deep learning*, and *natural language processing* reflect current AI-driven trends. Post-2020 entries, such as *COVID-19*, *circular economy*, and *digital transformation,* indicate responses to global disruptions. In 2023-2024, emerging technologies such as *large language models*, *blockchain*, and *cybersecurity* suggest a forward-looking research agenda. The diversity and evolution of keywords underscore the dynamic nature of engineering research indexed in SciELO, spanning traditional disciplines, applied technologies, and societal challenges.

3.5 Mapping knowledge structures

3.5.1 Core journals and direct citation clusters

Fig. 5 presents the citation network of engineering journals indexed in SciELO, clustered using the Louvain algorithm. In bibliometric analysis, a citation network is a graphical representation in which nodes correspond to documents (e.g., journals or articles) and edges represent citation links between them ([13]). These networks help identify thematic clusters, influential sources, and patterns of knowledge dissemination across disciplines. The Louvain clustering revealed 14 distinct clusters, each grouping journals with stronger internal citation ties. Each resulting cluster is depicted in a different color, with node size reflecting keyword frequency and edge width corresponding to the number of co-occurrences between terms. The cluster, most densely populated by grouping 13 sources, includes highly cited journals, such as DYNA, INF TECNOL, and INTERCIENCIA, suggesting this cluster forms the core of SciELO's engineering literature. These journals exhibit high publication volume and citation impact, indicating centrality in the network. The journals BOL MALARIOL SALUD AMBIENT contain the second central cluster, J TECHNOL MANAGE IINOV, and REV LASA INVEST. REV ING CONSTR, INGENIERIA, and REV ALCONPAT conform to the third most crucial cluster. There are no more essential relationships to highlight among the journals.

3.5.2 Intellectual structure of Latin American engineering research

The co-citation network [14] (**Fig. 6**), constructed from references to journals in SciELO that are also indexed in Scopus, reveals the intellectual structure underpinning Latin American engineering research. The analysis shows that the network is organized into three distinct clusters, each representing a group of journals that are frequently cited together, suggesting thematic or disciplinary affinities. The first cluster includes journals such as REV CONSTR, COMPUT SIST, and J APPL RES TECHNOL, focusing on applied engineering, construction, and computing technologies. This cluster reflects interdisciplinary connections within practical and technological domains. The second cluster is centered around REV MEX ING QUIMICA and ELECTRON J BIOTECHNOL, suggesting a strong emphasis on chemical engineering and biotechnology, with CTyF CIENC TECNOL FUTURO contributing additional coverage of scientific and technological development themes. The third comprises DYNA, INGEN INVEST, and ING UNIVERS, three general engineering journals that are among the most prolific in the region. Their co-citation is central to supporting foundational engineering knowledge and cross-cutting topics. The presence of distinct, well-formed clusters reflects the specialization of Latin American engineering journals. At the same time, their internal cohesion suggests the formation of stable, recognizable thematic communities within the regional research ecosystem.

Figure 3. Dominant author keywords per year
Source: The authors.

3.6 Thematic clustering of SciELO engineering journals

Fig. 4 presents a correlation map of SciELO engineering journals based on the cleaned author keywords, which serve as the cross-variable. The size of each node is proportional to the number of documents published by the corresponding journal. In this map, node colors correspond to clusters detected by the Louvain algorithm, where each color represents a group of journals sharing stronger internal keyword correlations than with the rest of the network. At the same time, the width and intensity of the connecting lines reflect the strength of the cross-correlations between journals. The map reveals a well-defined core of closely interrelated journals, all actively publishing in engineering fields. This core includes journals such as DYNA, BOLETIN TECNICO, and COMPUT SIST. In contrast, journals like TERRA LATINOAM and PAP PHYS appear on the periphery of the correlation map, suggesting that their thematic focus diverges from the core journals' central topics. The map highlights a strong correlation between ANAL INVEST

ARQUIT and ESTOA, suggesting a shared thematic scope or overlapping research interests within a specific subfield.

3.6.1 Thematic affinities

A journal coupling network is a bibliometric structure in which links between journals are established based on the number of shared references they cite [15], [16], [17]. The more two journals cite the same sources, the stronger their coupling relationship. This method reveals similarities in the intellectual foundations and thematic orientations of journals, regardless of whether they cite each other directly. This study constructed the coupling network using journals indexed in both SciELO and Scopus. The Louvain algorithm identified four clusters, each representing a distinct thematic community based on shared citations.

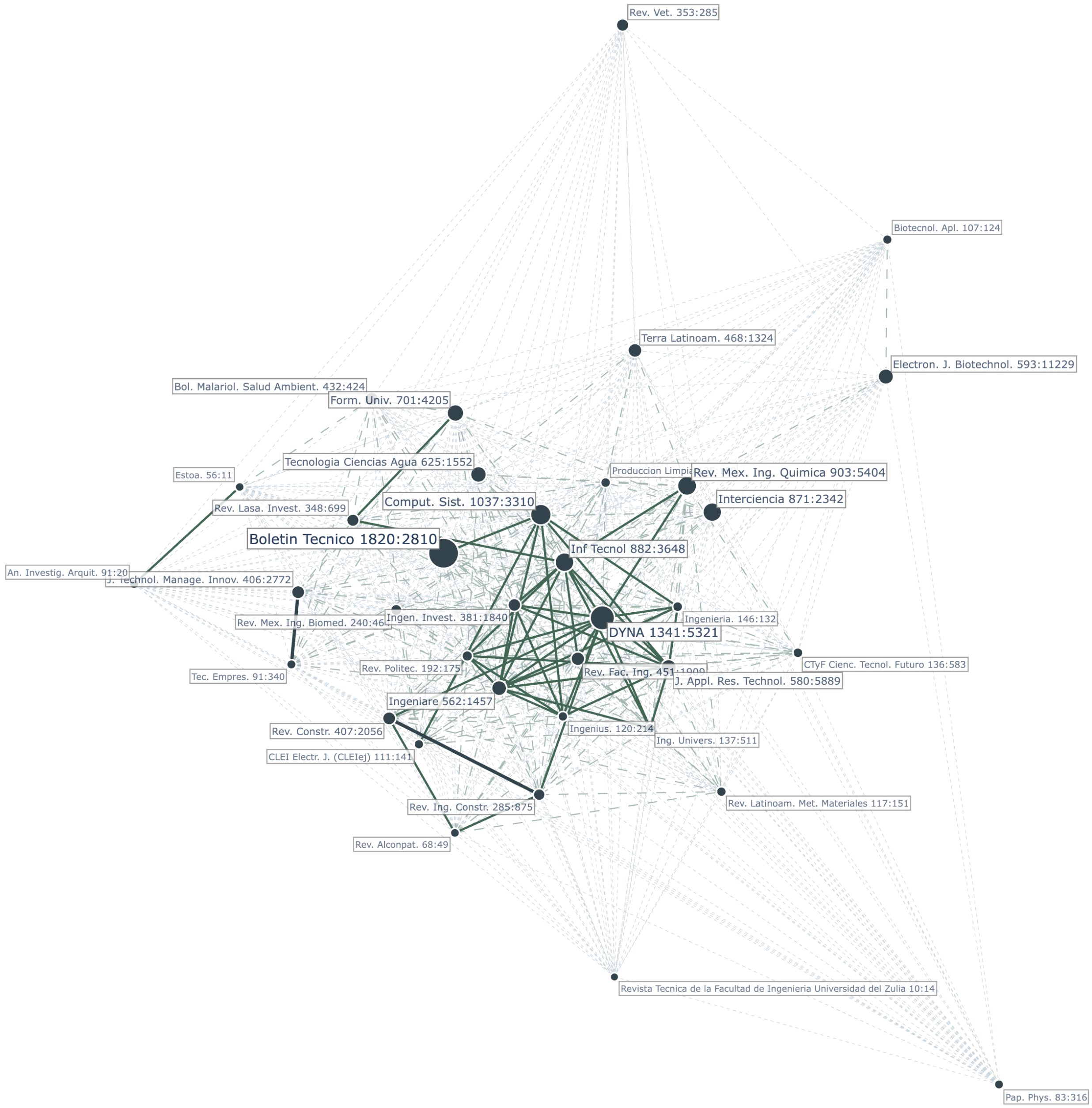

Figure 4. Correlation map of the engineering SciELO journals based on the author keywords.
Source: The authors.

Fig. 7 presents the obtained coupling network, which contains four clusters. In the coupling network, each color corresponds to a cluster of journals with similar reference patterns, and edge thickness indicates the strength of bibliographic coupling between journals. The first cluster, the largest, includes general and applied engineering journals such as FORM UNIV, J APPL RES TECHNOL, COMPUT SIST, and INGEN INVEST, reflecting a core of publications grounded in multidisciplinary engineering education, technology, and construction. The second cluster features journals like INF TECHNOL, INTERCIENCIA, and REV FAC ING, suggesting a group focused on innovation, institutional research, and technology management. The third cluster brings together journals such as REV MEX ING QUIMICA, ELECTRON J BIOTECHNOL, and TERRA LATINOAM, indicating a strong thematic alignment with chemical engineering, biotechnology, and environmental sciences. The fourth cluster group, comprised of CTYF CIENC TECNOL FUTURO and DYNA, forms a tightly coupled pair, possibly representing a more interdisciplinary or policy-oriented focus. The coupling network reveals clear thematic divisions and shared intellectual foundations among Latin American engineering journals.

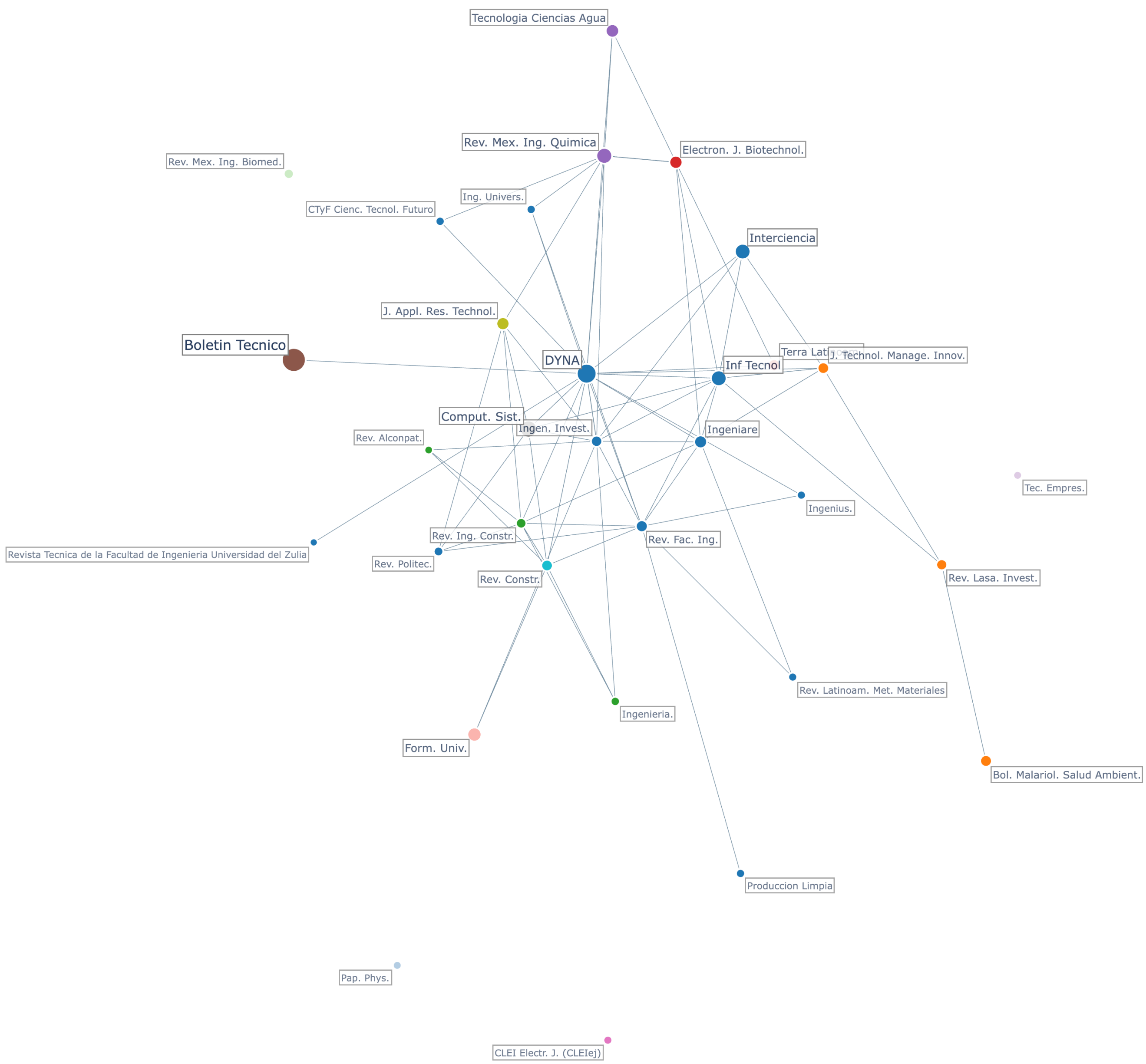

Figure 5. Citation network.
Source: The authors.

3.7 Dominant Themes

Table 5 presents the eight main dominant themes identified in engineering journals indexed in SciELO. These themes were obtained by applying the Louvain algorithm to the co-occurrence network of author keywords. A co-occurrence network is a graph in which nodes represent keywords, and links connect keywords that appear together in the same document. The strength of the link increases with the number of co-occurrences, revealing relationships between concepts and uncovering thematic structures. For this analysis, only cleaned author keywords with a frequency of five or more occurrences were included, ensuring the focus remained on the most relevant and recurring terms across the dataset. The Louvain algorithm, a widely used community detection method in network analysis, groups densely connected nodes into clusters, allowing for the identification of coherent topics or research areas. Each resulting cluster represents a dominant theme within the SciELO engineering literature. This method offers a data-driven approach to thematic mapping, highlighting the field's conceptual organization and providing insights into how topics are interlinked across publications.

The eight dominant themes identified through this analysis are described in detail below, highlighting their internal coherence, conceptual focus, and relevance within the broader landscape of engineering research in Latin America.

Similar thematic structures have been reported in bibliometric studies from fields such as environmental sciences, computer science, and health research, where keyword co-occurrence networks and thematic clusters also reveal discipline-wide intellectual communities and emerging research fronts.

3.7.1 Bioremediation and bioenergy for environmental sustainability

This thematic cluster integrates biogenic materials, fermentation processes, and waste valorization to address pollution, enhance water quality, and generate renewable products. Studies highlight the transformation of biomass into antimicrobial and antioxidant compounds, such as copper nanoparticles from bacterial polysaccharides [15] and nanoemulsions from essential oils [16]. Fermentation emerges as a key process for producing high-value food and beverage products with enhanced biochemical stability and antioxidant capacity ([17], [18]). Wastewater treatment through biosorption and microbial reduction effectively removes heavy metals and endocrine disruptors [19], [20]. Simultaneously, civil engineering studies employing rice husk ash and silica fume emphasize circular strategies for reducing environmental impact [21]. These works reveal a systemic shift toward eco-innovative bioprocesses that integrate food, energy, and material systems, positioning biomass not as residue but as a central agent of bioremediation, sustainability, and renewable energy production [22], [23], [24].

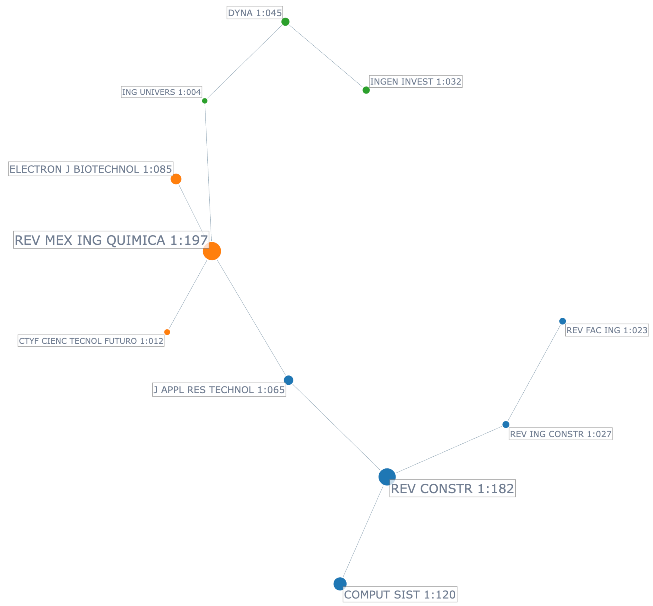

Figure 6. Co-citation network
Source: The authors.

3.7.2 Innovation, ICT, and knowledge management in higher education

This thematic cluster reflects the convergence of technological innovation, entrepreneurship, and knowledge management as transformative forces in higher education. Universities are increasingly integrating Information and Communication Technologies (ICTs) into teaching, management, and performance enhancement, thereby fostering adaptive learning and institutional efficiency [25], [26], [27]. Entrepreneurship emerges as a strategic objective linked to the development of small and medium enterprises (SMEs), supported by academic-industry collaboration and student-driven initiatives [25], [28], [29]. These dynamics align with values-based education and social responsibility, preparing students for socio-environmental challenges through digital competencies and civic engagement [30]. As institutions embrace digital transformation, universities evolve into responsive platforms for innovation, sustainability, and socio-economic development [31], [32], [33].

3.7.3 AI and data-driven technologies

This thematic cluster highlights the convergence of artificial intelligence, machine learning, and data-driven technologies applied to complex classification and optimization tasks across domains such as healthcare, agriculture, engineering, and digital security. Core methods, such as convolutional neural networks, support vector machines, and fuzzy logic, are used to extract and interpret high-dimensional data, thereby enhancing prediction accuracy and decision-making efficiency [25], [34], [35], [36]. Applications emphasize the development of context-aware, scalable systems enabled by big data, cloud computing, and the Internet of Things (IoT) infrastructures [26], [27]. Energy efficiency and sustainability emerge as underlying priorities in computational model design [34], [37]. Natural language processing extends these approaches to unstructured data, illustrating AI's cross-functional versatility [27]. Collectively, these studies reflect a shift toward intelligent, adaptive systems that unify digital architectures and learning-based models to solve real-world engineering problems [25], [38].

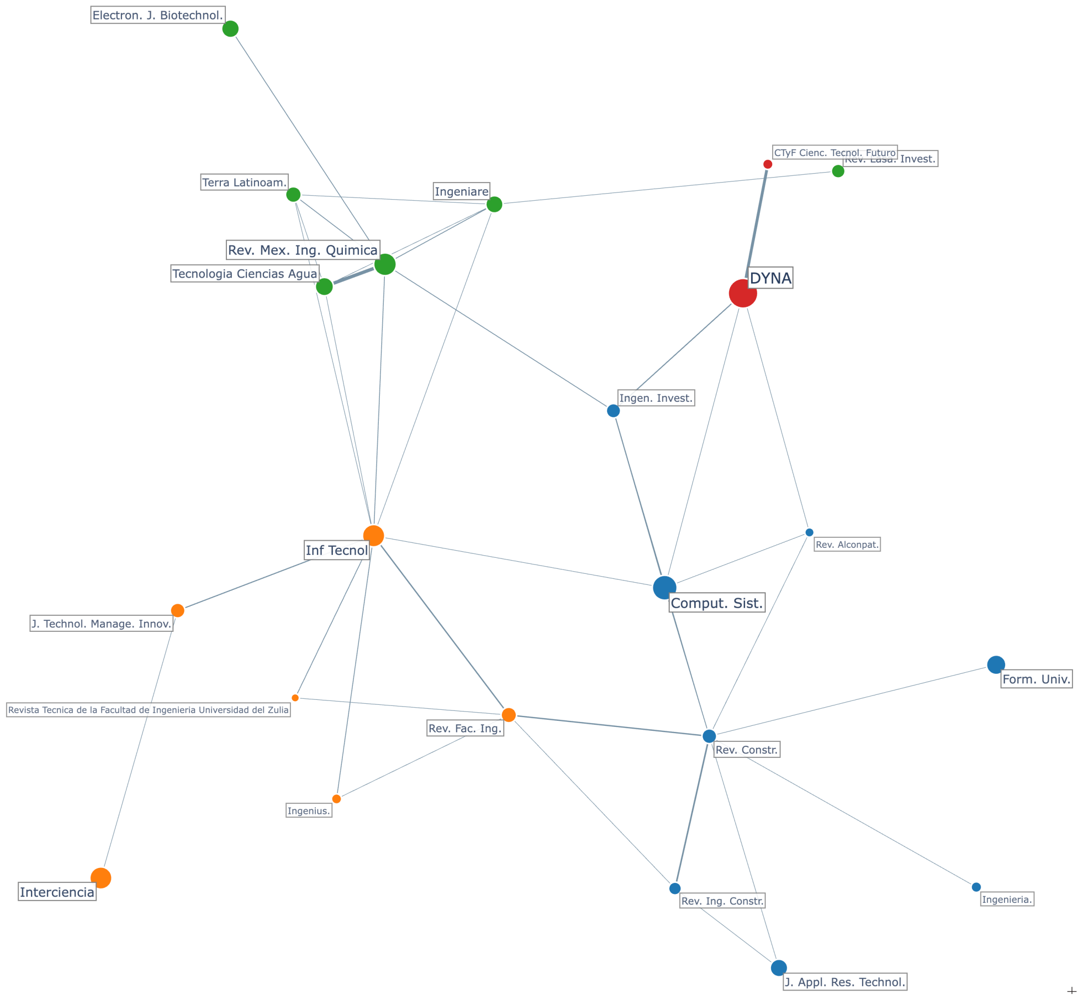

Figure 7. Coupling network.
Source: The authors.

Table 5.
Dominant themes.

Theme Name	Num Terms	Percentage	Main Terms
Bioremediation and bioenergy for environmental sustainability	478	24.8 %	Adsorption; biomass; antioxidant; heavy metals; fermentation; antioxidant activities; wastewater; pollution; biofuel; bio diesel
Innovation, ICT, and knowledge management in higher education	413	21.4 %	Higher education; innovations; education; university; information and communication technology; learning; small and medium enterprises; management; university students; knowledge management
AI and data-driven technologies	339	17.6 %	Artificial neural network; machine learning; data mining; big data; deep learning; support vector machine; natural language processing; artificial intelligence; internet of things; convolutional neural network
Engineering materials and structural analysis	290	15.0 %	Finite element analysis; mechanical property; concretes; computational fluid dynamics; nano particle; compressive strength; numerical simulations; corrosion; chitosan; rheology
Climate modeling and risk assessment	130	6.7 %	Simulations; modeling; temperatures; climate change; remote sensing; precipitation; mathematical modeling; risks; accessibility; risks assessment
Optimization and intelligent engineering systems	111	5.8 %	Optimization; genetic algorithm; response surface methodology; reliability; micro grid; wireless sensor network; multi objective optimization; power quality; heuristics; production
Epidemiology and public health	103	5.3 %	Covid19; dengue; pandemic; risk factors; dogs; prevalence; cattle; epidemiology; stresses; prevention
Sustainability, circular economy, and environmental management	64	3.3 %	Sustainability; sustainable development; recycling; circular economy; solar energies; environmental impacts; agriculture; environmental education; solid wastes; costs

Source: The authors.

3.7.4 Engineering materials and structural analysis

This thematic cluster reflects the convergence of material science, environmental engineering, and computational modeling to enhance mechanical performance, durability, and sustainability in engineering applications. Studies emphasize the use of agricultural and industrial residues—such as fly ash, chitosan, and coffee husk ash—to improve concrete and composite materials while supporting waste valorization and pollutant removal [19], [24], [39]. Numerical simulations, including finite element analysis and computational fluid dynamics, guide structural optimization and microstructural analysis under diverse loading and environmental conditions [40], [41]. Rheological assessments and kinetic models contribute to understanding degradation processes and thermal stability, especially in systems incorporating nano- and bioactive components [28], [42]. Applications range from predicting concrete strength using image analysis [43] to studying the transformation of endocrine disruptors [44], illustrating a shift toward multifunctional, simulation-driven, and eco-efficient material solutions grounded in experimental validation and predictive modeling.

3.7.5 Climate modeling and risk assessment

This thematic cluster focuses on integrating mathematical modeling, simulations, and geospatial analysis to assess environmental vulnerabilities driven by climate variability and anthropogenic pressures. Studies use remote sensing and geostatistical tools to simulate the impacts of precipitation, temperature, and pollution on water systems, informing risk assessments and sustainability strategies [45], [46], [47], [48]. Emphasis is placed on evaluating system behavior under dynamic conditions, including water accessibility and air quality degradation, often using empirical data to guide scenario-based interventions [20], [23], [49]. Climate models reveal how slight temperature or precipitation shifts substantially alter hydrological flows, highlighting the compounded effects of climate change [48]. Accessibility and environmental justice also intersect with sustainability objectives, connecting technical evaluations to social outcomes [29]. Overall, the cluster advances a systems-level framework for understanding and managing environmental risk through simulation, spatial analysis, and predictive modeling.

3.7.6 Optimization and intelligent engineering systems

This thematic cluster reflects the integration of multi-objective optimization and metaheuristic algorithms with energy systems, production processes, and infrastructure management to enhance reliability, efficiency, and sustainability. Techniques such as genetic algorithms and response surface methodology are widely applied in smart grids, microgrids, and wireless sensor networks to optimize power quality, distributed generation, and system robustness under uncertainty [16], [19], [25]. These methods support predictive maintenance and energy-efficient production, particularly in scenarios that require real-time decision-making and adaptive control [37], [38]. The convergence of algorithmic models with practical applications—ranging from pH optimization in environmental remediation to dynamic response tuning in control systems—underscores a systems-oriented approach that blends computational intelligence with sustainable engineering design [21], [50].

3.7.7 Epidemiology and public health

This thematic cluster reflects a multidisciplinary approach to zoonotic and vector-borne diseases, emphasizing the convergence of epidemiological surveillance, environmental stressors, and digital communication to enhance public health resilience. Across the abstracts, infectious diseases, such as COVID-19, dengue, malaria, and canine-related illnesses, are studied for their prevalence, risk factors, and spatial dynamics in both human and animal populations, suggesting a One Health framework [15], [16], [51]. The role of social media in risk communication and mental health detection during pandemics demonstrates a shift toward digital epidemiology [25], [52], [53]. Studies also highlight the use of modeling and indices to forecast disease patterns and support prevention strategies under various climatic and urban conditions [54], [55], [56]. The integration of engineering approaches—ranging from environmental modeling to biomedical innovations—positions this cluster at the intersection of disease ecology, technological adaptation, and health systems preparedness.

3.7.8 Sustainability, circular economy, and environmental management

This thematic cluster reflects a systems-oriented integration of environmental sustainability, circular economy, and technological innovation across engineering and agricultural domains. Studies emphasize the reuse of materials such as rice husk ash, refractory waste, and coffee husk ash to reduce the environmental footprint of concrete while preserving its mechanical performance [39], [47]. Food engineering research supports sustainability through drying and fermentation processes that enhance nutritional value and improve energy efficiency by utilizing agro-industrial residues [23], [57]. Simultaneously, biosurfactant-based pollutant control [19] and biotechnological innovations such as copper nanoparticles and essential oil nanoemulsions [15], [16] reinforce low-impact solutions. The roles of life cycle assessment, social responsibility, and environmental education emerge in efforts to align industrial processes, university practices, and consumer behavior with sustainability goals [58], [59], positioning engineering research within a multidimensional ecological framework.

4 Conclusions

This study provides a multidimensional portrait of engineering journals from Latin America indexed in both SciELO and Scopus, revealing a complex interplay between structural visibility, thematic specialization, and regional collaboration. The cross-correlation of Scopus subject areas reveals a broad but uneven disciplinary coverage, with clusters of journals focusing on applied fields, including civil, environmental, and industrial engineering. Performance indicators highlight a concentration of scientific output among a small group of institutions and countries, underscoring the presence of regional research hubs. Thematic analysis of author keywords reveals recurring areas, including sustainability, materials science, and energy, and highlights inconsistencies in terminological usage across journals. Correlation networks based on keywords and citations reveal both thematic proximities and intellectual communities, with co-citation and bibliographic coupling patterns indicating shared epistemic foundations. Using keyword co-occurrence analysis, we identified eight dominant themes that reflect the diversity and priorities of engineering research in Latin America, encompassing topics such as sustainability, public health, data-driven technologies, and core engineering domains. The findings confirm SciELO's strategic role in enhancing regional visibility and suggest that journals with stronger thematic focus and inter-journal connectivity tend to achieve greater integration in the global scientific landscape. Ultimately, this bibliometric mapping highlights the evolving structure and knowledge production patterns of Latin American engineering research, providing valuable guidance for scholars, journal editors, and policymakers seeking to strengthen regional scientific communication and enhance the international relevance of their work.

For authors, the results highlight where thematic opportunities and visibility gaps exist across journals, helping align research topics with regional and global impact areas. For editors, the structural and thematic maps provide actionable insights to refine journal scope, strengthen interdisciplinary positioning, and enhance citation connectivity. For policymakers, the findings offer evidence to guide investment in emerging research fronts, support regional collaboration networks, and reinforce SciELO's strategic role in scientific capacity-building.

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